

Optimizing Culling Strategies in Preventing the Spread of Chronic Wasting Disease in Wild Deer Populations

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Background

Chronic Wasting Disease (CWD) is a fatal prion disease that affects white-tailed deer and other North American cervids. CWD is caused by the incorrect folding of proteins called prions, which will result in the progressive destruction of brain tissue. CWD is spread by both direct contact with infected individuals and contact with contaminated soil, food, or water [1]. More specifically, infection can occur from contact with infected saliva, blood, urine, or carcasses [2]. Infected animals can appear healthy for months or years, while at the same time transmitting the disease to other animals. Symptoms of CWD include weight loss, poor coordination, drooling, frequent urination, and drooping ears. No treatments or vaccines exist for CWD [2]. As of 2025, CWD has been found in at least 36 U.S. states since it was first reported in the 1960s, and is expected to continue spreading [1].

One of the most difficult challenges in preventing CWD is the role that an environmental prion reservoir plays in maintaining the prevalence of CWD. Prions that are shed from infected deer can bind to soil, plants, and water while remaining infectious for years [3]. This means that even if the infected population has been effectively eliminated, the disease can still infect through interaction with the contaminated environment. As such, once CWD has been established in an area, these reservoirs make it "extremely difficult to manage and eliminate the disease" [3].

Because no treatment exists, many hunters and wildlife managers utilize culling as a tool for reducing the number of infected animals, which should slow down the spread of the infection and, by extension, the development of these reservoirs [3]. However, due to the early, asymptomatic nature of CWD, most (but not all) culling is non-targeted, meaning infected and non-infected deer are removed from the population indiscriminately. It is not until the deer's remains are processed at regional facilities that it is confirmed whether or not they were infected [3]. As such, culling can be described as a tradeoff; too little culling and the disease is not being effectively prevented, too much and the wild deer population suffers from over-hunting.

The objective of this project is to answer this question: **What is the optimal culling rate that minimizes the prevalence of CWD while keeping the deer population above 75% of its disease-free carrying capacity?**

This question is important because, from the perspective of a wildlife manager, understanding the optimal strategy for culling provides direct insight into managing the aforementioned tradeoff; a model that can provide a clear culling rate (proportional to the population) gives authorities a quantitative basis for how to effectively cull the population while maintaining ecologically sustainable levels.

We can construct a compartmental ODE model that represents the susceptible, exposed, and infected deer populations alongside an environmental prion reservoir. This model will incorporate logistic growth, direct and environmental transmission, congenital transmission, and non-targeted culling. We will calculate the basic reproduction number R_0 using the next generation matrix approach, as well as use MATLAB software to identify an optimal value for the culling parameter. Then, we will analyze how environmental conditions affect the optimal culling rate.

Assumptions and Notation

First, we assume our population is homogeneous, meaning every deer can equally interact with any other deer. We also assume that the wild deer population follows a basic, logistic growth with some carrying capacity K , so the population has its own internal birth and death process. In this model, we do not consider any cause of death other than natural, CWD-induced, or culling (meaning we exclude predators, accidents, etc.).

Reproduction occurs in only two compartments; susceptible and exposed. Susceptible deer reproduce normally and the offspring are born into susceptible. The sources noted that CWD can be transmitted congenitally [4] (i.e. mother-to-offspring), which is reflected in the exposed reproduction terms; a fraction, denoted by p , of the exposed offspring are born into exposed, and the rest $(1 - p)$ will enter susceptible. It is reasonable to assume that deer in the advanced stages of CWD are not capable of supporting viable offspring, so we will assume there is no reproduction in the infected compartment. All reproduction follows a logistic growth pattern.

Deer exist in only one of three compartments: susceptible (S), exposed (E), or infectious (I). As stated previously, there is no recovery, CWD is completely fatal, so there exist no paths back to susceptible once infected. The fourth compartment (P) represents the concentration of the prion in the environment. We assume that only infected deer "feed" the environmental reservoir via shedding prions. Prions in the environment will eventually decay at some rate δ .

As shown in the sources, deer can transmit the disease through direct deer-to-deer contact or through environmental contact. When a susceptible deer is infected, it transfers from susceptible to exposed, and then finally transfers to infectious at some rate σ . This represents the asymptomatic incubation period mentioned in the background. Infected deer will eventually die and leave the system at some rate α . For simplicity, we are going to assume that exposed deer do not significantly contribute to environmental or direct transmissions.

Culling occurs immediately at $t = 0$. Culling acts equally at a constant rate γ_c in all three deer compartments (S, E, I) because it is non-targeted.

The parameters are as follows. All parameters take positive values and all populations are non-negative.

Parameter	Description
r	Growth rate of deer population
K	Carrying capacity of deer population
μ	Natural death rate
β_d	Direct transmission rate (i.e. deer-to-deer infection)
β_e	Environmental transmission rate (i.e. contaminated environment to deer)
σ	Progression rate from exposed to infectious populations
α	CWD-induced death rate
ω	Prion shedding rate (i.e. rate of contaminating environment)
δ	Prion natural decay rate
γ_c	Culling rate (i.e. fraction of total population removed per unit time)
p	Probability of congenital transmission

Finally, we will quantify our goals. Our first goal is to utilize culling to minimize the prevalence of CWD in the deer population, which means minimizing the ratio $\frac{I^*}{N^*}$, where I^* and N^* represent the steady states of the two populations over a long period of time. Our second goal is to ensure that, after culling, the total deer population N stays within three-quarters of its carrying capacity K . Written quantitatively, these goals become:

- (1) Minimize $\frac{I^*}{N^*}$
- (2) Maintain the inequality $N^* \geq 0.75K$

The ODE system that we will use is as follows:

$\dot{S}$	$rS \left(1 - \frac{N}{K}\right) + (1-p)rE \left(1 - \frac{N}{K}\right) - \beta_d SI - \beta_e SP - \mu S - \gamma_c S$
$\dot{E}$	$\beta_d SI + \beta_e SP + prE \left(1 - \frac{N}{K}\right) - \sigma E - \mu E - \gamma_c E$
$\dot{I}$	$\sigma E - \alpha I - \mu I - \gamma_c I$
$\dot{P}$	$\omega I - \delta P$
N	$S + E + I$

For convenience, we will use the factored version going forward:

$\dot{S}$	$r(S + (1-p)E) \left(1 - \frac{N}{K}\right) - (\beta_d I + \beta_e P)S - (\mu + \gamma_c)S$
$\dot{E}$	$(\beta_d I + \beta_e P)S + prE \left(1 - \frac{N}{K}\right) - (\sigma + \mu + \gamma_c)E$
$\dot{I}$	$\sigma E - (\alpha + \mu + \gamma_c)I$
$\dot{P}$	$\omega I - \delta P$
N	$S + E + I$

Solution

Total Deer Population Equation

$$\dot{N} = \dot{S} + \dot{E} + \dot{I}$$

Looking at the compartments from the previous section, we can see that multiple terms cancel each other out. The transmission terms $\beta_d SI$ and $\beta_e SP$ will cancel out from the sum of susceptible and exposed. We can also see that σE will cancel as well. The congenital birth terms will combine to produce rE . We can factor out μ and γ_c from all compartments and substitute in N . We are left with:

$$\dot{N} = r(S + E)(1 - \frac{N}{K}) - \mu N - \alpha I - \gamma_c N$$

From this, we can see the deer population does not remain constant; it fluctuates with the birth and death terms present in our model.

Disease Free Equilibrium (DFE) using Next Generation Method

$$\text{Infectious Compartments} = \vec{x} = \begin{pmatrix} E \\ I \\ P \end{pmatrix} \quad \text{Non-Infectious Compartments} = \vec{y} = \begin{pmatrix} S \end{pmatrix}$$

When $\vec{x} = 0$, this implies that:

$$\dot{S} = rS(1 - \frac{S}{K}) - (\mu + \gamma_c)S$$

The DFE is therefore (assuming $S^* \neq 0$):

$$\begin{aligned} 0 &= S^*[r(1 - \frac{S^*}{K}) - (\mu + \gamma_c)] \\ 0 &= r(1 - \frac{S^*}{K}) - (\mu + \gamma_c) \\ r(1 - \frac{S^*}{K}) &= \mu + \gamma_c \\ \frac{S^*}{K} &= 1 - \frac{\mu + \gamma_c}{r} \\ S^* &= K(1 - \frac{\mu + \gamma_c}{r}) = K(\frac{r - \mu - \gamma_c}{r}) \end{aligned}$$

We can see that in order for the DFE to exist, the following condition must be satisfied:

$$r > \mu + \gamma_c$$

Calculating R_0 - the basic reproduction number

In this context, R_0 represents the number of secondary infections caused by a single deer infected with CWD introduced to the susceptible population.

$$F_E = \beta_d SI + \beta_e SP + prE(1 - \frac{N}{K})$$

$$V_E = (\sigma + \mu + \gamma_c)E$$

$$F_I = 0$$

$$V_I = (\alpha + \mu + \gamma_c)I - \sigma E$$

$$F_P = 0$$

$$V_P = \delta P - \omega I$$

$$F(S^*) = \left[\frac{\partial F_i}{\partial x_j} \right] (S^*) = \begin{bmatrix} pr(1 - \frac{S^*}{K}) & \beta_d S^* & \beta_e S^* \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} p(\mu + \gamma_c) & \beta_d S^* & \beta_e S^* \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$V = \left[\frac{\partial V_i}{\partial x_j} \right] = \begin{bmatrix} \sigma + \mu + \gamma_c & 0 & 0 \\ -\sigma & \alpha + \mu + \gamma_c & 0 \\ 0 & -\omega & \delta \end{bmatrix}$$

Using MATLAB (See Appendix-1), we get the following after calculating the eigenvalues of FV^{-1} :

$$\begin{aligned} R_0 &= \frac{\delta p \gamma_c^2 + 2\delta p \gamma_c \mu + \alpha \delta p \gamma_c + \delta p \mu^2 + \alpha \delta p \mu + S^* \beta_d \delta \sigma + S^* \beta_e \omega \sigma}{\delta(\alpha + \gamma_c + \mu)(\gamma_c + \mu + \sigma)} \\ R_0 &= \frac{\delta p(\gamma_c^2 + 2\gamma_c \mu + \mu^2 + \alpha \gamma_c + \alpha \mu) + S^* \sigma(\beta_d \delta + \beta_e \omega)}{\delta(\alpha + \gamma_c + \mu)(\gamma_c + \mu + \sigma)} \\ R_0 &= \frac{\delta p((\gamma_c + \mu)^2 + \alpha(\gamma_c + \mu)) + S^* \sigma(\beta_d \delta + \beta_e \omega)}{\delta(\alpha + \gamma_c + \mu)(\gamma_c + \mu + \sigma)} \\ R_0 &= \frac{\delta p(\gamma_c + \mu)(\gamma_c + \mu + \alpha) + S^* \sigma(\beta_d \delta + \beta_e \omega)}{\delta(\alpha + \gamma_c + \mu)(\gamma_c + \mu + \sigma)} \\ R_0 &= \frac{\delta p(\gamma_c + \mu)(\gamma_c + \mu + \alpha)}{\delta(\alpha + \gamma_c + \mu)(\gamma_c + \mu + \sigma)} + \frac{S^* \sigma(\beta_d \delta + \beta_e \omega)}{\delta(\alpha + \gamma_c + \mu)(\gamma_c + \mu + \sigma)} \\ R_0 &= \frac{p(\gamma_c + \mu)}{\gamma_c + \mu + \sigma} + \frac{S^* \sigma(\beta_d \delta + \beta_e \omega)}{\delta(\alpha + \gamma_c + \mu)(\gamma_c + \mu + \sigma)} \end{aligned}$$

Theorem check from March 24th lecture

Condition 1: $F_i(0, \vec{y}) = 0 \quad V_i(0, \vec{y}) = 0 \quad \forall y_i \geq 0$

Set $E = I = P = 0$ and substitute to validate:

$$F_E = \beta_d S(0) + \beta_e S(0) + pr(0)(1 - \frac{N}{K}) = 0$$

$$V_E = (\sigma + \mu + \gamma_c)(0) = 0$$

$$F_I = 0$$

$$V_I = (\alpha + \mu + \gamma_c)(0) - \sigma(0) = 0$$

$$F_P = 0$$

$$V_P = \delta(0) - \omega(0) = 0$$

Condition 2: $\vec{y}' = g(0, \vec{y})$ has a unique locally asymptotic stable equilibrium.

The disease-free equilibrium has already been calculated: $S^* = K(\frac{r - \mu - \gamma_c}{r})$

Condition 3: $F_i(\vec{x}, \vec{y}) \geq 0 \quad \forall x_i, y_i \geq 0$

$$F_I = 0 \geq 0$$

$$F_P = 0 \geq 0$$

$$F_E = \beta_d SI + \beta_e SP + prE(1 - \frac{N}{K})$$

The term $prE(1 - \frac{N}{K})$ is problematic because it's plausible that $N > K$ (i.e. current total population is above the carrying capacity), which could return a negative value for F_E . As such, condition 3 fails. To be thorough, we will continue checking the remaining conditions.

Condition 4: $V_i(x, y) \leq 0$ whenever $x_i = 0$.

For $E = 0$

$$V_E = (\sigma + \mu + \gamma_c)(0) = 0 \leq 0$$

For $I = 0$

$$V_I = (\alpha + \mu + \gamma_c)(0) - \sigma E = -\sigma E \leq 0$$

For $P = 0$

$$V_P = \delta(0) - \omega I = -\omega I \leq 0$$

Condition 5: $\sum_{i=1}^k V_i(x, y) \geq 0 \quad \forall x_i, y_i \geq 0$

$$\begin{aligned} \sum V_i &= V_E + V_I + V_P \\ \sum V_i &= (\sigma + \mu + \gamma_c)E + (\alpha + \mu + \gamma_c)I - \sigma E + \delta P - \omega I \\ \sum V_i &= (\mu + \gamma_c)E + \delta P + (\alpha + \mu + \gamma_c - \omega)I \end{aligned}$$

The first two terms must be non-negative because they are the sum and product of positive or non-negative terms.

The third term is problematic due to its coefficient: $(\alpha + \mu + \gamma_c - \omega)$. It is possible for $\omega > \alpha + \mu + \gamma_c$, which could cause the overall sum to be negative if $E = P = 0$.

This means that the theorem does not hold if the prion shedding rate exceeds the combined effect of deer leaving the system through natural or unnatural death.

Conditions 3 and 5 are not satisfied, and the theorem cannot be applied to our system. We cannot conclude whether or not S^* is stable or unstable using the theorem. Instead, we may use MATLAB simulations in the following section to visually determine whether our disease-free equilibrium S^* is stable.

MATLAB Simulations

We are going to run multiple MATLAB simulations in order to better understand what the trajectories of each compartment look like over time. By varying parameters such as the culling rate, we can empirically observe which choice of parameters produces better results concerning our stated goals.

4.0.1 Baseline Trajectories

The very first simulation we will run is plotting the trajectories of each compartment when there is no active culling ($\gamma_c = 0$). This is done to get a sense of how the populations evolve without human intervention.

We begin by choosing reasonable values for our parameters as follows:

r	K	μ	β_d	β_e	σ	α	ω	δ	γ_c	p
0.6	1000	0.1	0.0005	0.00025	0.5	0.8	1	0.1	0	0.5

Using MATLAB (see Appendix.2), we can plot the trajectories of the compartments to produce the following graph:

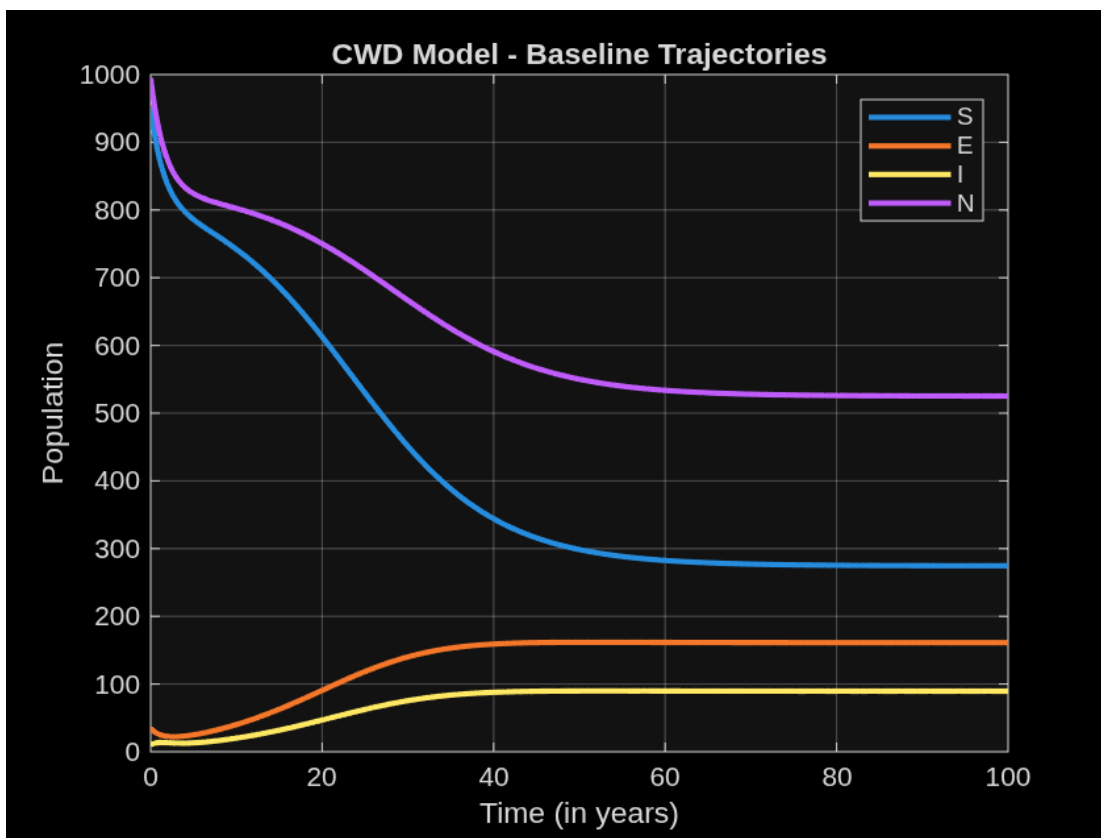


Figure 1: No Culling - Baseline Trajectories

4.0.2 Gradually increasing the culling rate

Next, let's gradually increase the culling rate to see how the total population and the infected population react to culling. We can graph these populations separately to see how culling impacts their long-term trajectories. The MATLAB code (See Appendix-3) produces the following two graphs:

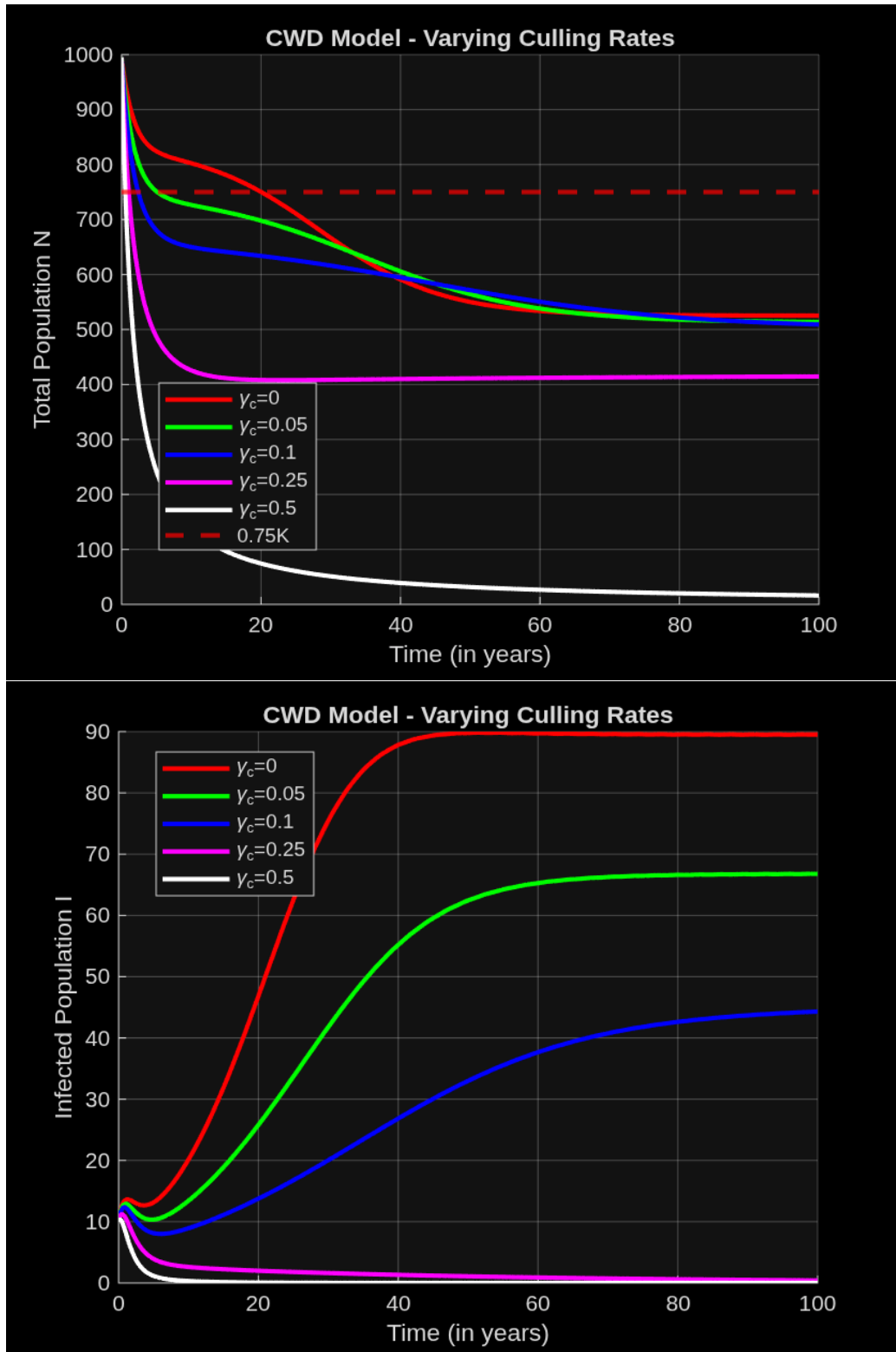


Figure 2: Total population (Top) and infected population (Bottom) for various culling rates.

A quick observation is that the optimal culling value lies somewhere roughly between $\gamma_c = 0.1$ and $\gamma_c = 0.25$. This is the threshold where we see the infected population dying off, hence meeting goal 1. However, we also observe that even with no culling present, the population naturally decreases towards a long-term value of around 500 deer, well below our goal of 750 deer.

Now that we have a rough idea of the optimal γ_c , let's iteratively increase γ_c in small step sizes and see what the infection prevalence and overall long-term population become. (See Appendix-4)

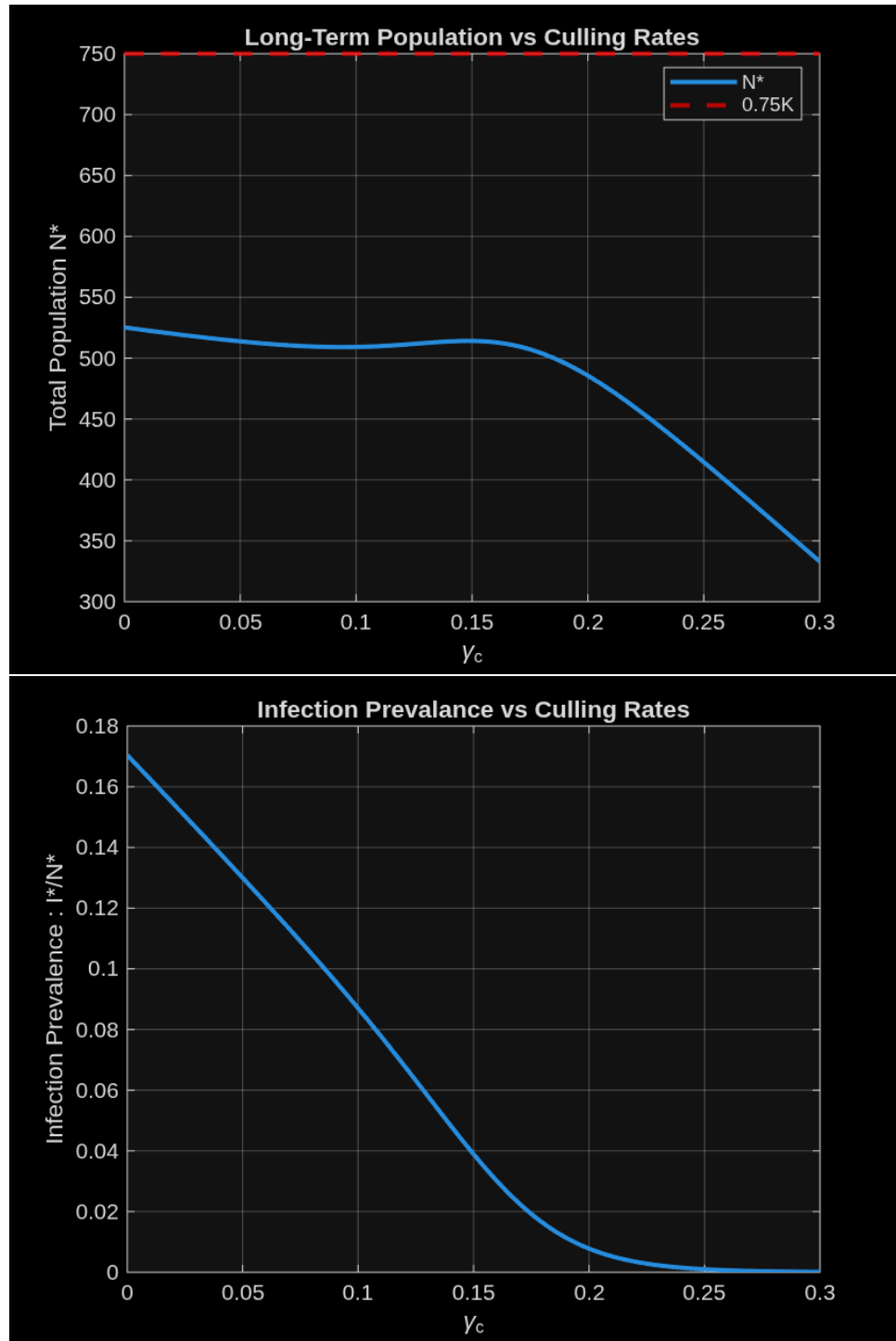


Figure 3: Long term population (Top) and infection prevalence (Bottom) for various culling rates.

4.0.3 Scenario: Increasing the prion decay rate (i.e. What happens if prions are removed from the environment at a higher rate?)

We will repeat the previous step, but this time we will see what happens when $\delta = 0.25$ and $\delta = 0.5$.

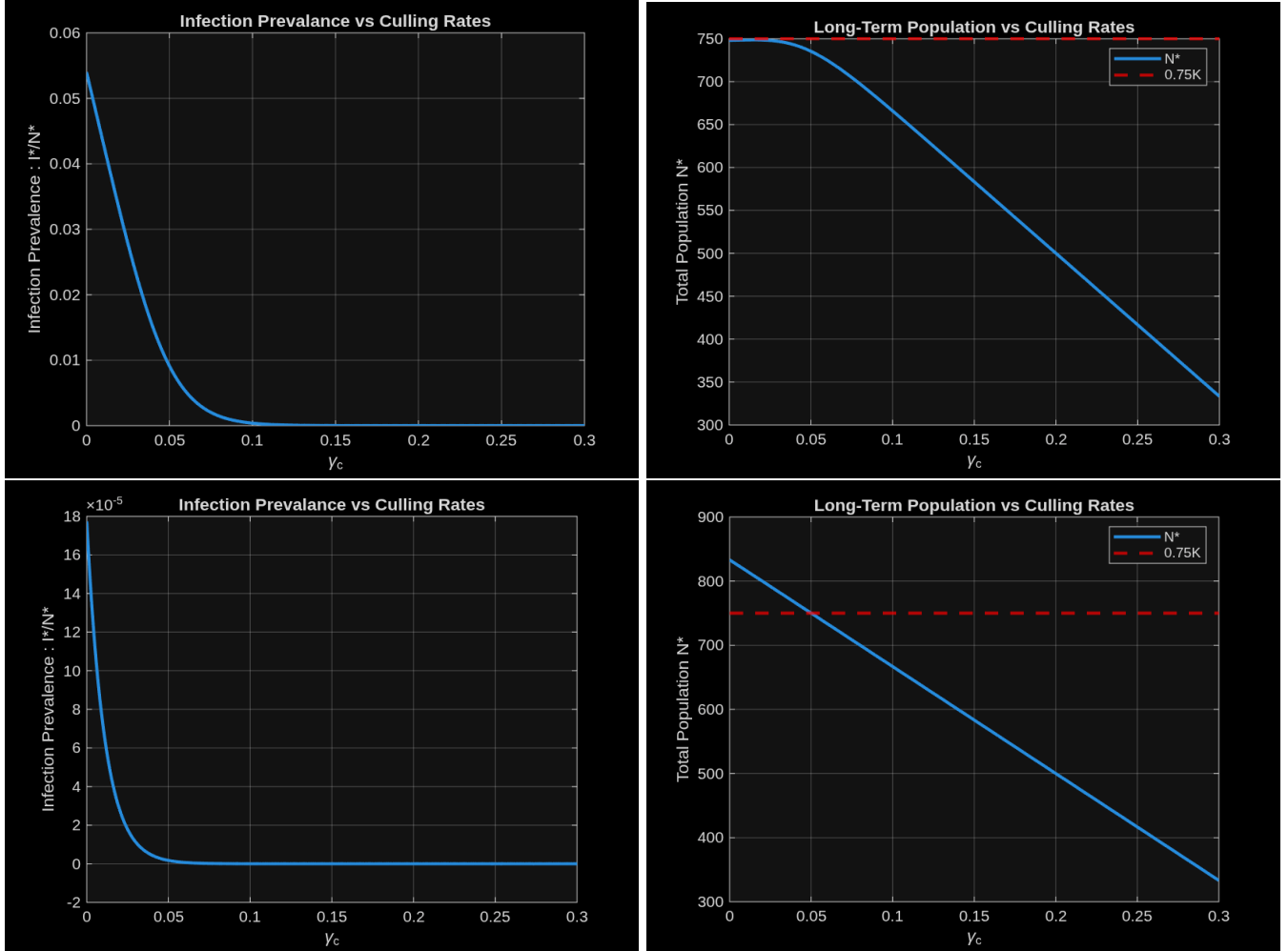


Figure 4: Infection prevalence (left) and long-term population (right) for $\delta = 0.25$ (top) and $\delta = 0.5$ (bottom).

A quick observation shows that increasing δ will clearly decrease the infected prevalence while at the same time increasing the long-term population; thus satisfying both goals at once.

In fact, in the case of $\delta = 0.5$, a culling rate of $\gamma_c \approx 0.05$ will result in a nearly eradicated infected population while the overall population remains just above the $0.75K$ threshold. This contrasts with the previous step, where we were unable to achieve both goals regardless of what value the culling rate took.

4.0.4 Scenario: Altering the balance of transmission parameters (i.e., What if the environment was more infectious than direct transmission, vice-versa)

We are going to repeat step 2, but now we are going to swap the transmission parameters to see how this affects the infection prevalence and long-term population.

Our default assumption was that the environmental transmission was weaker than direct transmission. Now, we are going to reverse that assumption and assume that the environment can infect deer at a higher rate than direct transmission. Using a decay rate of $\delta = 0.25$, an environmental transmission rate of $\beta_e = 0.0005$, and a direct transmission rate of $\beta_d = 0.00025$, we get the following graphs:

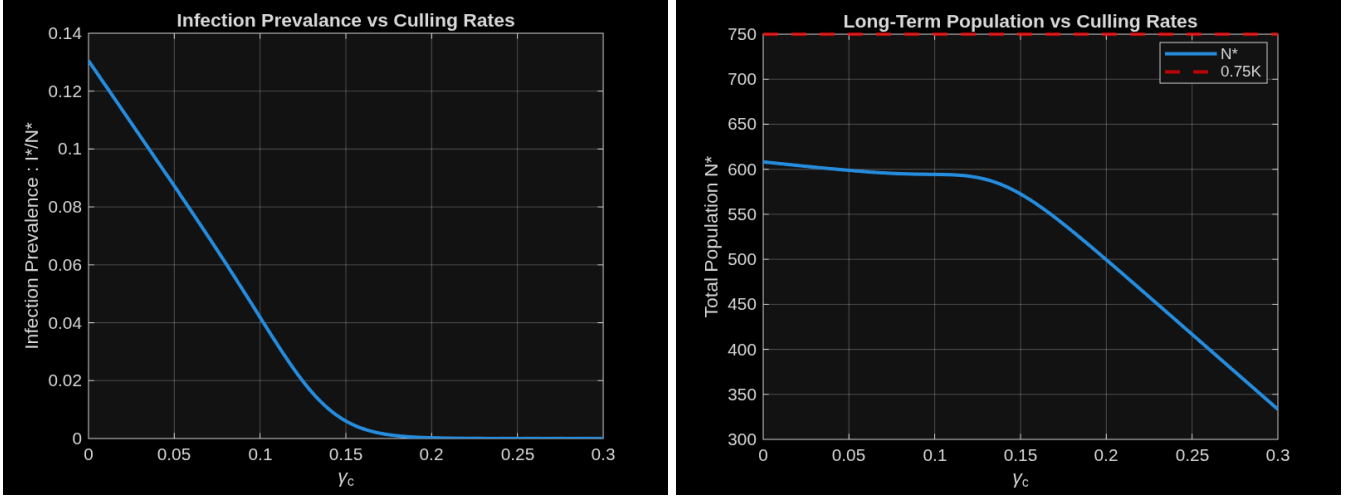


Figure 5: Swapped Transmission - Infection prevalence (left) and long-term population (right) for $\delta = 0.25$

Compare this to our graphs for $\delta = 0.25$ from the previous section:

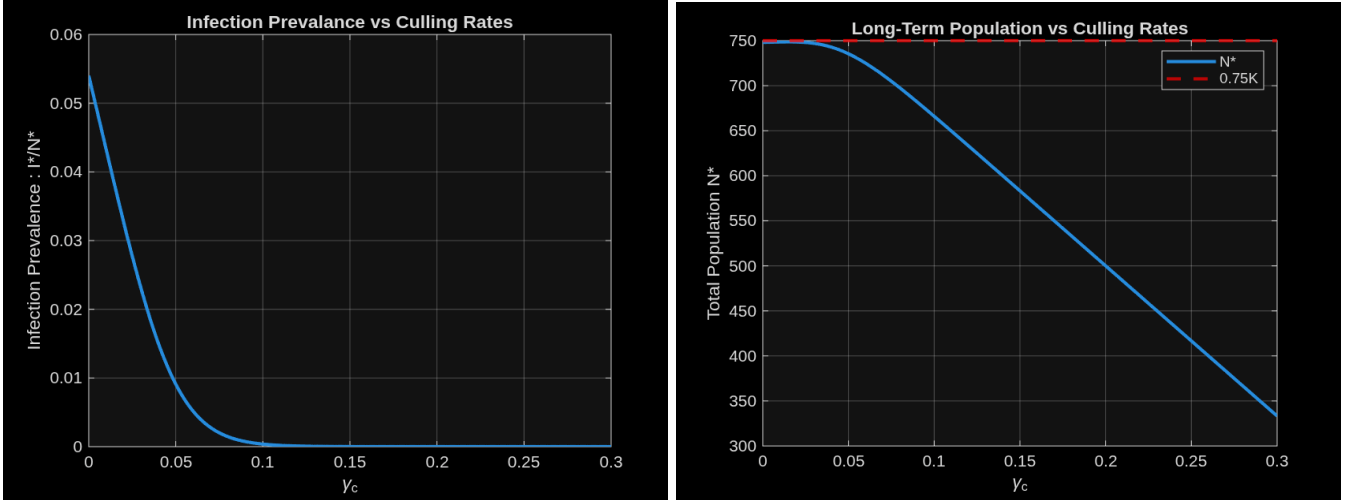


Figure 6: Infection prevalence (left) and long-term population (right) for $\delta = 0.25$

For lower values of γ_c , we can observe that the swapped model has a lower overall population than the default setup, but for higher culling values, they are roughly equal. Also, we observe that we require more aggressive culling ($\gamma_c \approx 0.2$) to achieve infection minimization compared to the default setup ($\gamma_c \approx 0.1$).

Interpreting Results

From observing Figures 1, 2, and 3, we can conclude that no culling rate will ever satisfy goal 2. Even when no culling is present, Figure 1 shows that the population will naturally decrease and converge to approximately 500 deer. However, we were able to meet goal 1; the infectious population was effectively eradicated at culling rates between 20-25% of the deer population, leaving an overall population of around 400-450 deer. Whether such a drastic reduction in the species is necessary to eradicate the disease is controversial; it's possible that allowing a certain percentage of the population to be infected is acceptable as long as the overall population is above a certain threshold.

Increasing δ , the prion decay rate, produced the most influential results in our simulation. As δ grew, we observed a lower prevalence of CWD and a higher long-term deer population, meaning both of our goals could be met. In fact, we observed (Figure 4) that for $\delta = 0.5$ and $\gamma_c = 0.05$, we were able to effectively eradicate CWD while maintaining a deer population of around 750, satisfying both goals. A culling rate of 5% is much more reasonable, but the question remains: How can we actually increase δ ? If we interpret the prion decay rate more generally as the rate at which prions leave the environment, it's possible that, alongside culling, environmental agencies could either strategically deploy prion-decaying chemical agents or physically remove contaminated parts of the environment, thus depleting the prion reservoir.

From Figures 5 and 6, we showed that the environmental transmission rate (β_e) is more influential in spreading CWD than through direct deer-to-deer transmission (β_d). When $\beta_e > \beta_d$, we observed that more aggressive culling rates were necessary to eradicate the disease. We also observed that, for lower culling rates, the long-term deer population was lower when compared with the default assumption that $\beta_e \neq \beta_d$. Combined with the previous point, this implies that the environmental prion reservoir is the main obstacle to preventing the spread of CWD, and thus, we should focus on removing the prion from the environment rather than focusing solely on culling the deer population.

Assumptions and Extraneous Factors

One of our first assumptions about the model was a homogeneous population. This assumption is somewhat inaccurate, as both geographic barriers and social behaviors will limit where a deer travels in its lifetime. Deer will typically gather at specific spots (e.g., grazing, water, shelter), rather than travel long distances without good reason. Deer also tend to engage in herd-like behavior, meaning that other groups of deer are unlikely to interact perfectly equally with each other. Introducing a spatial element to this model would make it more realistic.

Our model also ignored factors influencing mortality, such as the existence of local predators or accidents, such as vehicular collisions. It is reasonable to assume that a deer in the late stages of CWD would have a diminished capability to escape predators or evade accidents. As such, these factors could actually act as a sort of natural targeted-culling term, where infected deer are removed from the population because they no longer have the capability to survive against natural pressures in their environment.

One major assumption we made regarding transmission of CWD is that exposed deer do not contribute to the spread of CWD. Realistically, an exposed, asymptomatic deer is still capable of shedding prions into its environment, thus contributing to the environmental prion reservoir and making culling even less effective.

Finally, our choice of parameters for the simulations was selected based on biological plausibility (e.g. $r = 0.6$, meaning each deer produces 0.6 offspring on average). Instead, if we had access to real-life data regarding deer births and natural death rates, we could recalibrate the simulations to be more accurate. However, it's unlikely that this would significantly change the patterns we have already identified.

Summary

Under reasonable parameters, there does not exist any culling rate γ_c capable of minimizing the infected population while also keeping the total population above three-quarters of its carrying capacity. Culling alone is capable of eradicating the disease ($\gamma_c \approx 0.25$), but this will cause the total population to drop to ecologically controversial levels. However, when paired with increasing the decay rate of prions in the environment, it is possible to achieve both goals at a much more reasonable culling rate ($\gamma_c = 0.05$). In conclusion, the most effective strategy for combating CWD is a combination of mild culling and an environmental decontamination protocol, with emphasis on the latter.

Bibliography

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